Kwame Nkrumah University of
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ME 270
Manufacturing Technology

Lecture 5: Welding Processes

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Lecture Objectives

Understand the following:

- Arc Welding
- Resistance Welding
- Oxyfuel Gas Welding
- Other Fusion Welding Processes
- Solid State Welding
- Weld Quality
- Weldability
- Design Considerations in Welding

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Types of welding processes

- Fusion welding - coalescence is accomplished by melting the two parts to be joined, in some cases adding filler metal to the joint
- Examples: arc welding, resistance spot welding, oxyfuel gas welding
- Solid state welding - heat and/or pressure are used to achieve coalescence, but no melting of base metals occurs and no filler metal is added
- Examples: forge welding, diffusion welding, friction welding

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Arc Welding

- A fusion welding process in which coalescence of the metals is achieved by the heat from an electric arc between an electrode and the work
- Electric energy from the arc produces temperatures ~ 10,000 F (5500 C), hot enough to melt any metal
- Most AW processes add filler metal to increase volume and strength of weld joint

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Electric Arc

- An electric arc is a discharge of electric current across a gap in a circuit
- It is sustained by an ionized column of gas (*plasma*) through which the current flows
- To initiate the arc in AW, electrode is brought into contact with work and then quickly separated from it by a short distance

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Arc Welding

- A pool of molten metal is formed near electrode tip, and as electrode is moved along joint, molten weld pool solidifies in its wake

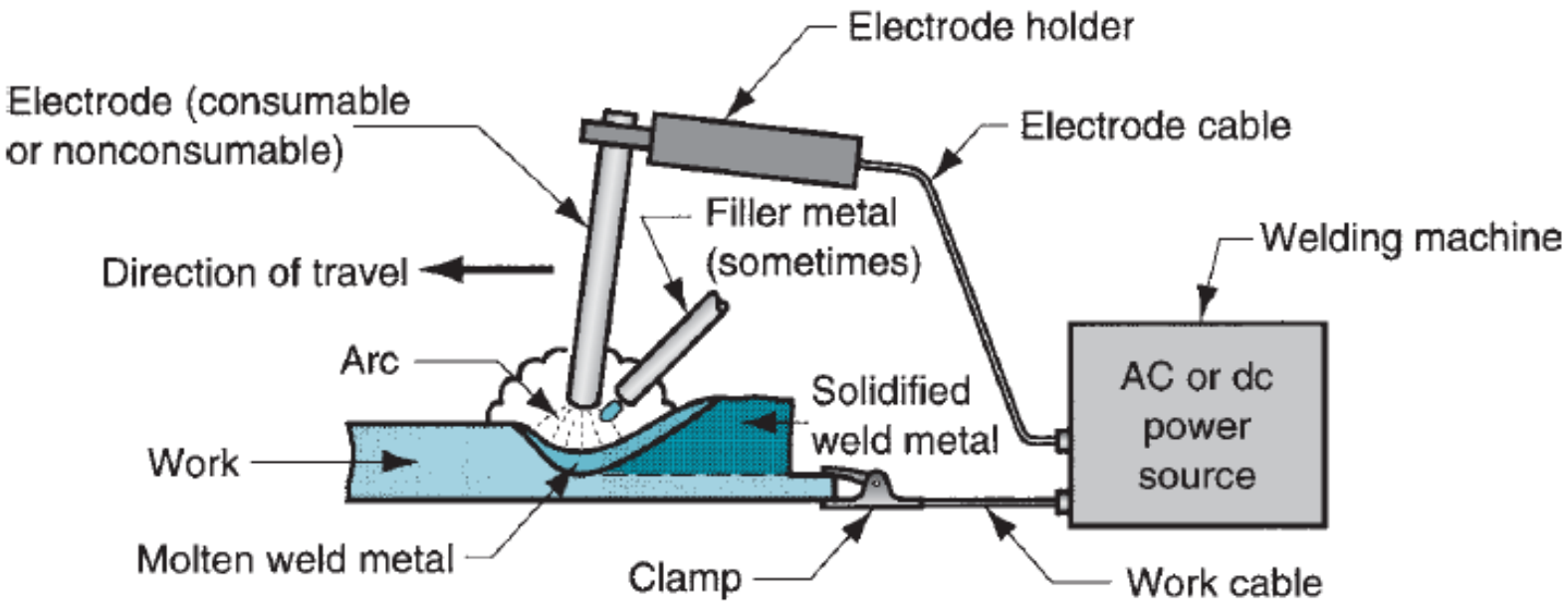


Fig. 1 Configuration of an arc welding

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Manual arc welding and arc time

- Problems with manual welding:

✓ Weld joint quality

✓ Productivity

$$\boldsymbol{Arc\ time = time\ arc\ is\ on / hours\ worked}$$

- Also called "arc-on time"
- Manual welding arc time = 20%
- Machine welding arc time ~ 50%

Arc welding electrodes

- There are two (2) types of arc welding electrodes:
- **Consumable electrodes** – They are consumed during welding process and provide the source of the filler metal in arc welding.
- **Non-consumable electrodes** – They are not consumed during welding process.

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Consumable electrodes

- Welding rods (also known as sticks) are 9 to 18 inches and 3/8 inch or less in diameter and must be changed frequently
- Weld wire can be continuously fed from spools with long lengths of wire, avoiding frequent interruptions
- In both rod and wire forms, electrode is consumed by the arc and added to weld joint as filler metal.
- Consumable welding rods must be changed periodically, reducing arc time of the welder

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Non-consumable electrodes

- They are not consumed during welding process because they are made of tungsten which resists melting.
- They gradually deplete during welding by vaporization.
- Filler metal is supplied by a separate wire fed into weld pool

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Arc Shielding

- At high temperatures in arc welding, metals are chemically reactive to oxygen, nitrogen, and hydrogen in air
- Mechanical properties of joint can be degraded by these reactions
- To protect operation, arc must be shielded from surrounding air in arc welding processes
- Arc shielding is accomplished by: Shielding gases, e.g., argon, helium, CO_2, and Flux.

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Power source in arc welding

- Both direct current (DC) and alternating current (AC) are used in arc welding.
- AC machines are less expensive to purchase and operate, but are restricted to welding of ferrous metals.
- DC machines can be used on all metals with good results and is generally noted for better arc control.

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Power balance in arc welding

- Power to drive the arc welding operation is the product of the current (I) passing through the arc and the voltage (E) across it.
- This power is converted into heat, but not all of the heat is transferred to the surface of the work due to losses through convection, conduction, radiation, and spatter.
- Power balance in arc welding is defined by

$$R_{HW} = f_1 f_2 IE = U_m A_w v$$

Where R_{HW} is rate of heat generated at the weld, f_1 is heat transfer factor, f_2 is melting factor, I is current, E is velocity, U_m is unit melting energy and v is volume melted.

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Example

A gas tungsten arc-welding operation is performed at a current of 300A and voltage of 20V. The melting factor f_2 = 0.5, and the unit melting energy for the metal Um=10 J/mm^3. Determine

(a) power in the operation

(b) rate of heat generation at the weld, and

(c) volume rate of metal welded.

[Take heat transfer factor as 0.7]

Solution

(a) The power in this arc-welding operation is

$P = IE = (300A)(20\ V) = 6000W$

(b) The rate of heat used for welding is given by

$R_{HW} = f_1 f_2 IE = (0.7)(0.5)(6000) = 2100W = 2100\ J/s$

(c) The volume rate of metal welded is

$R_{VW} = 2100/10 = 210\ mm^3/s$

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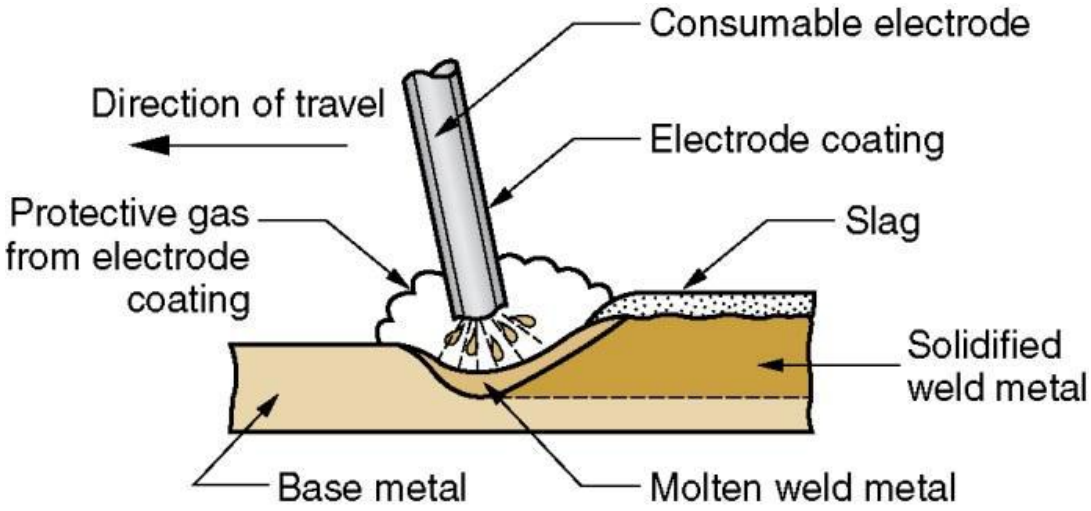
Consumable electrode arc welding processes

- Shielded Metal Arc Welding
- Gas Metal Arc Welding
- Flux-Cored Arc Welding
- Electrogas Welding
- Submerged Arc Welding

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Shielded metal arc welding

- Uses a consumable electrode consisting of a filler metal rod coated with chemicals that provide flux and shielding
- Sometimes called "stick welding"
- Power supply, connecting cables, and electrode holder available for a few thousand dollars



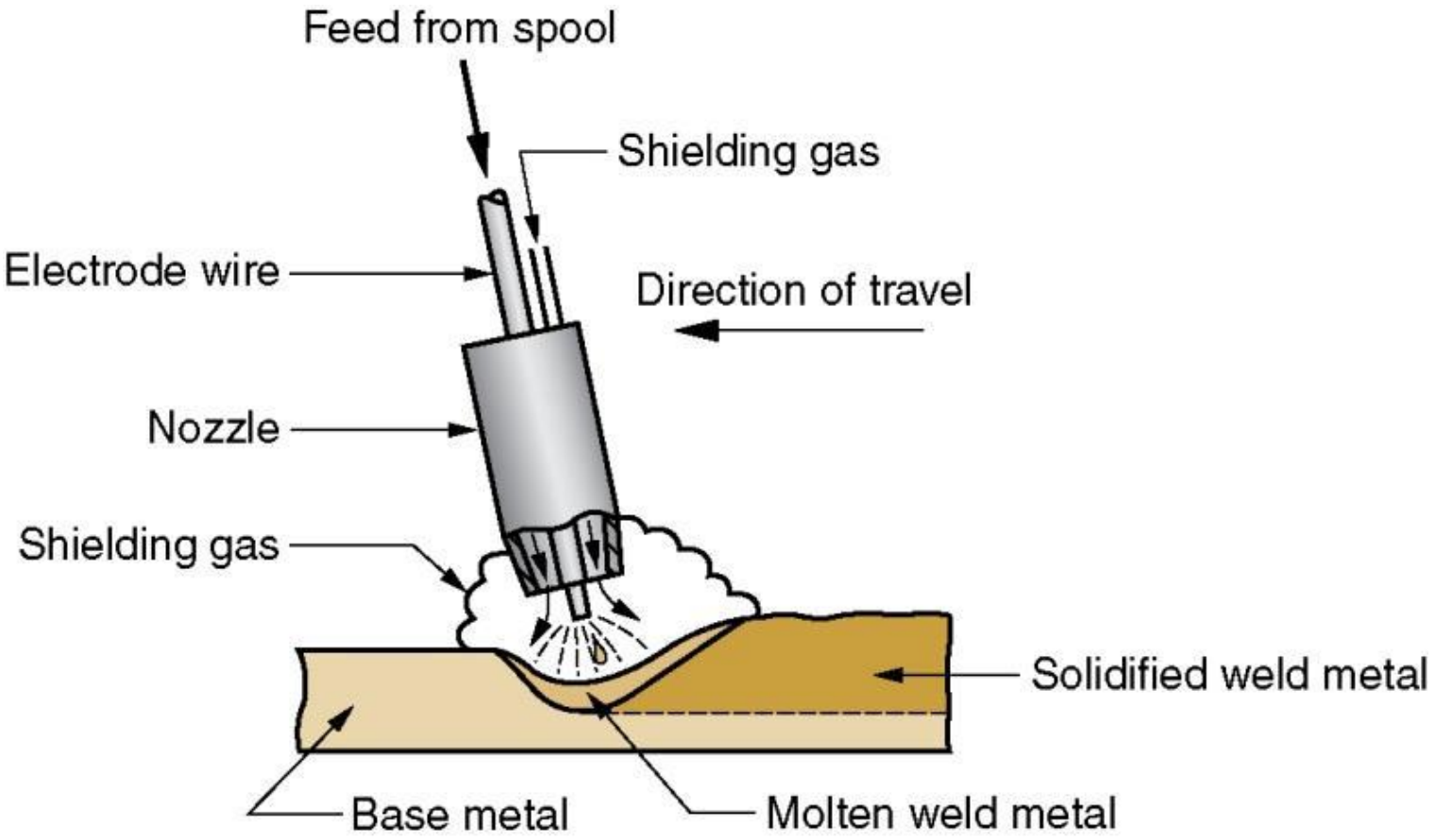
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Gas metal arc welding

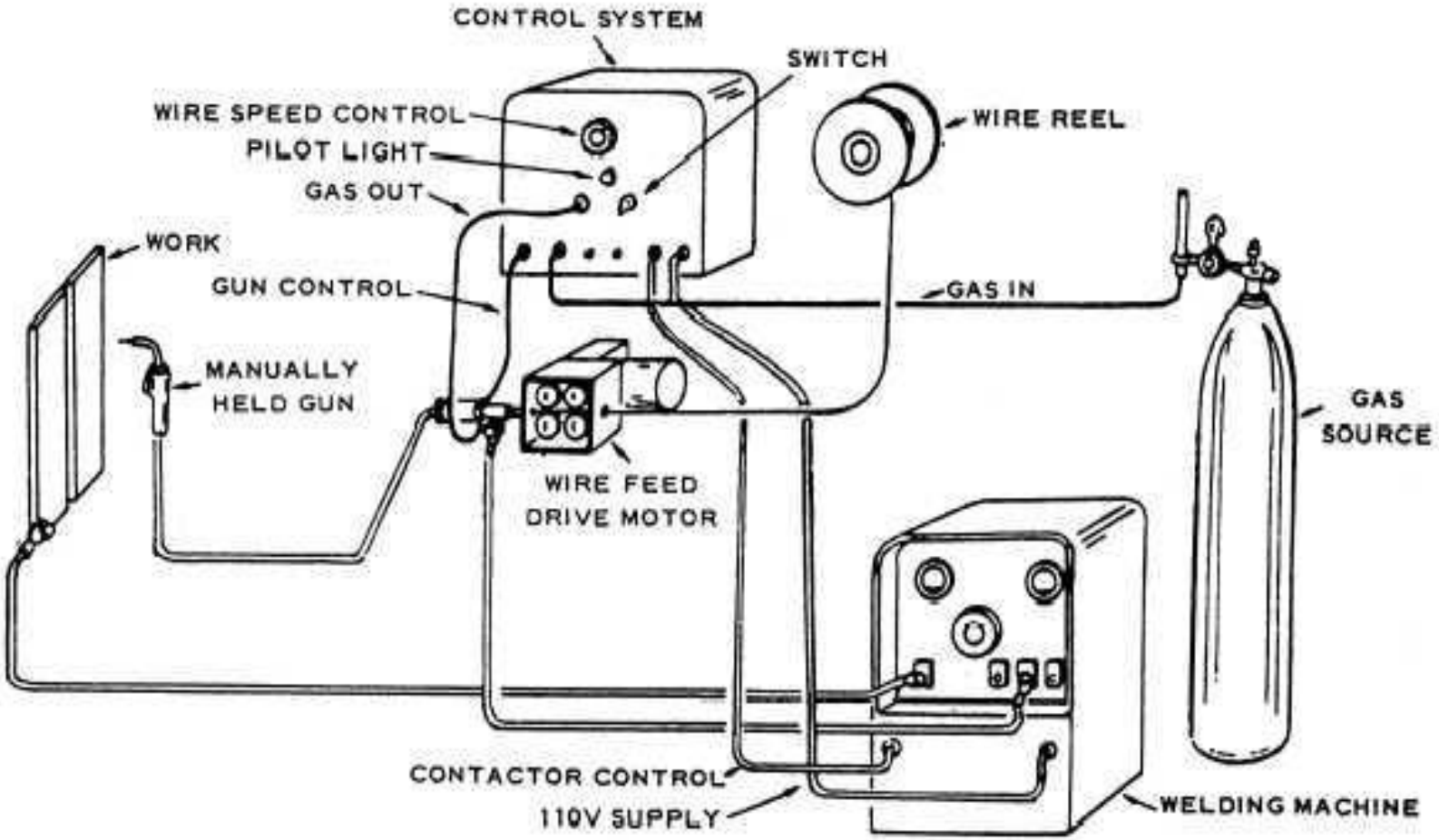
- Uses a consumable bare metal wire as electrode with shielding by flooding arc with a gas
- Wire is fed continuously and automatically from a spool through the welding gun
- Shielding gases include argon and helium for aluminium welding, and CO_2 for steel welding
- Bare electrode wire plus shielding gases eliminate slag on weld bead
- No need for grinding and cleaning of slag

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Gas metal arc welding



Gas metal arc welding



Advantages of gas metal arc welding over shield metal arc welding

- Better arc time because of continuous wire electrode
- Sticks must be periodically changed in SMAW
- Better use of electrode filler metal than SMAW
- End of stick cannot be used in SMAW
- Higher deposition rates
- Eliminates problem of slag removal
- Can be readily automated

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Flux-Cored Arc Welding (FCAW)

- Adaptation of shielded metal arc welding, to overcome limitations of stick electrodes
- There are two versions:
  - Self-shielded FCAW - core includes compounds that produce shielding gases
  - Gas-shielded FCAW - uses externally applied shielding gases
- Electrode is a continuous consumable tubing (in coils) containing flux and other ingredients (e.g., alloying elements) in its core.

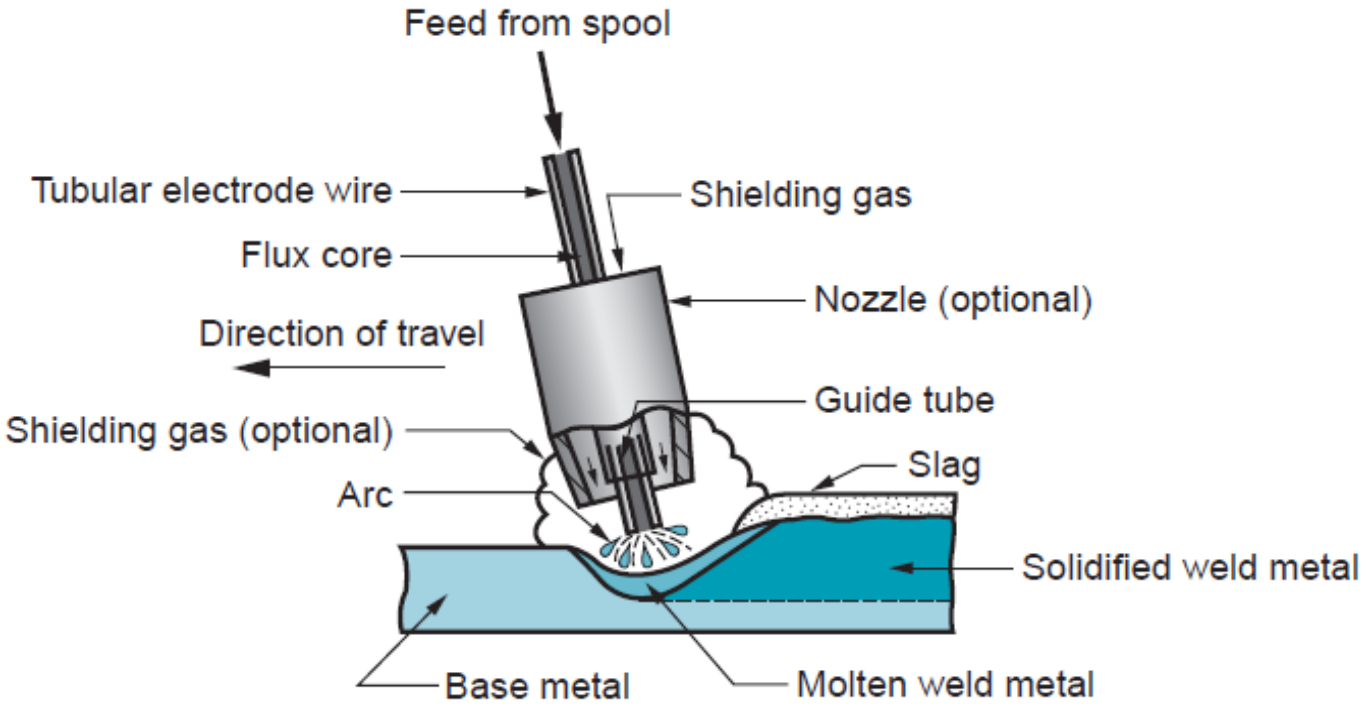
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Flux-Cored Arc Welding (FCAW)

- The presence or absence of externally supplied shielding gas distinguishes the two types:

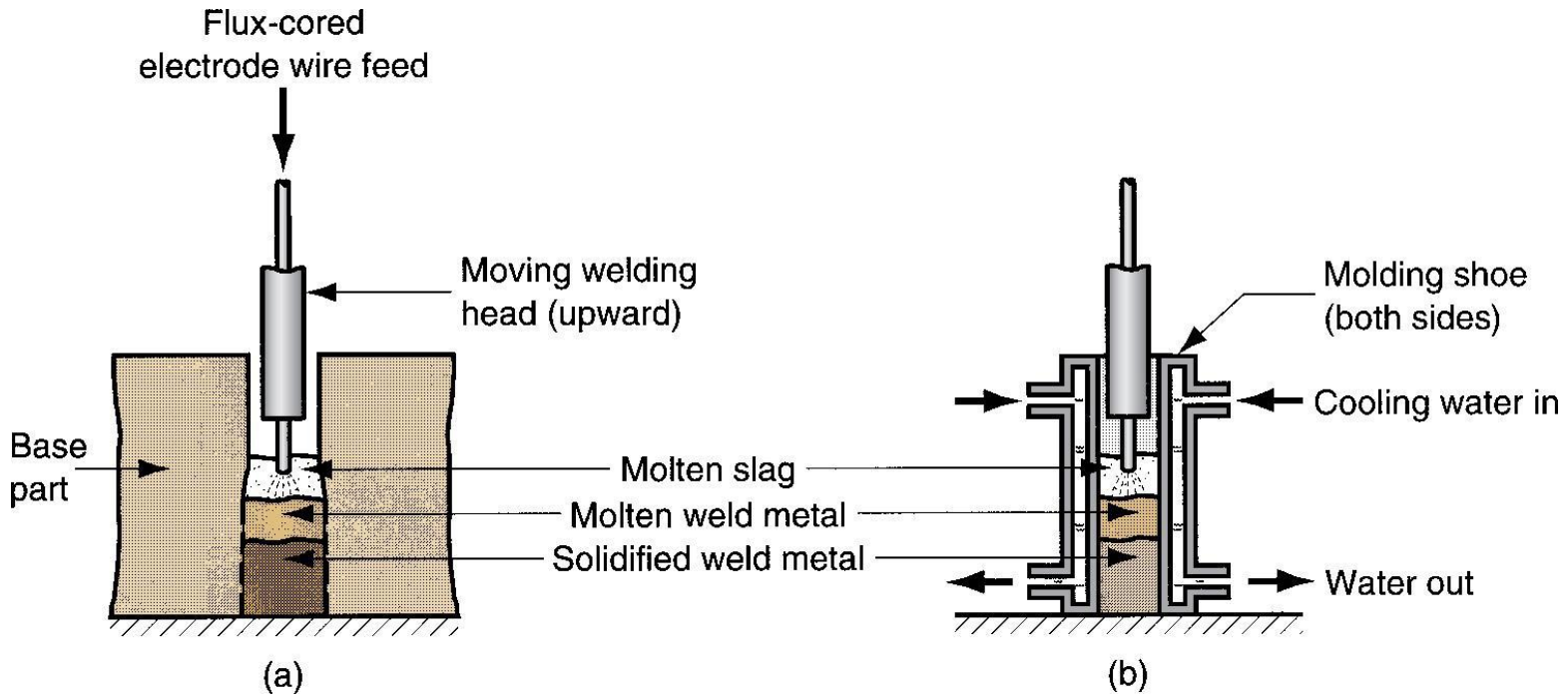
(1) **self-shielded**: The core provides the ingredients for shielding

(2) **Gas-shielded**: External shielding gases are supplied.



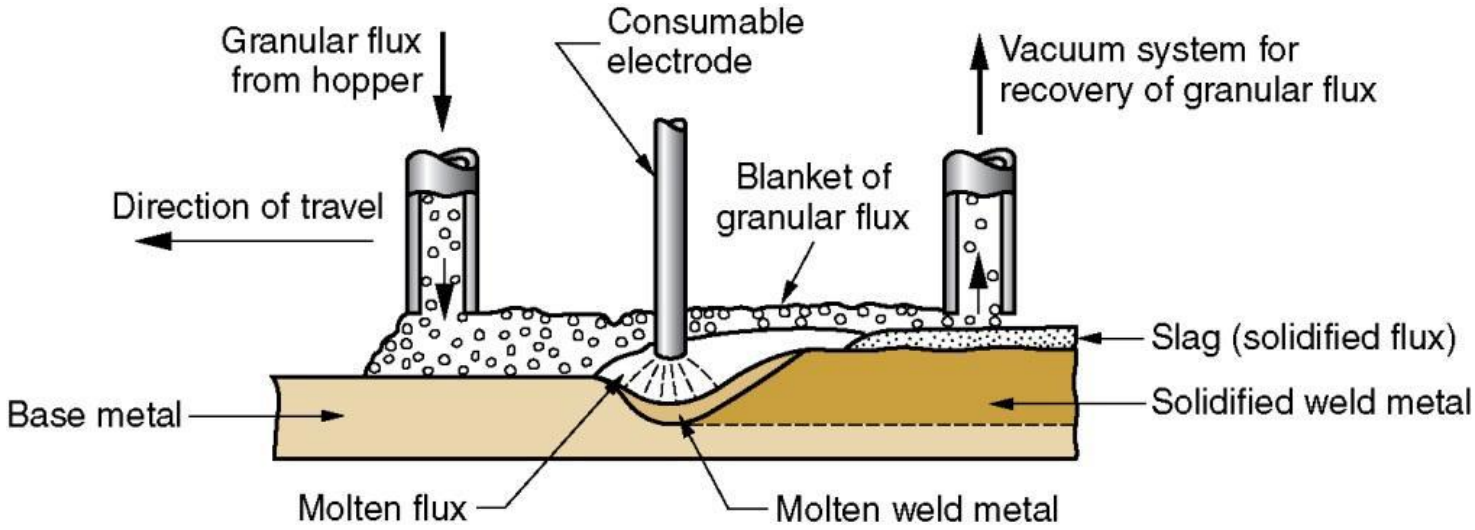
Electro-gas Welding

- Electro-gas welding (EGW) is an arc-welding process that uses a continuous consumable electrode (either flux-cored wire or bare wire with externally supplied shielding gases) and molding shoes to contain the molten metal.



Submerged Arc Welding (SAW)

- Uses a continuous, consumable bare wire electrode, with arc shielding by a cover of granular flux
- Electrode wire is fed automatically from a coil
- Flux introduced into joint slightly ahead of arc by gravity from a hopper
- Completely submerges operation, preventing sparks, spatter, and radiation.



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Applications od submerged arc welding

- Steel fabrication of structural shapes (e.g., I-beams)
- Seams for large diameter pipes, tanks, and pressure vessels
- Welded components for heavy machinery
- Most steels (except hi C steel)
- Not good for nonferrous metals

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Non-consumable electrode processes

- Gas Tungsten Arc Welding
- Plasma Arc Welding
- Carbon Arc Welding
- Stud Welding

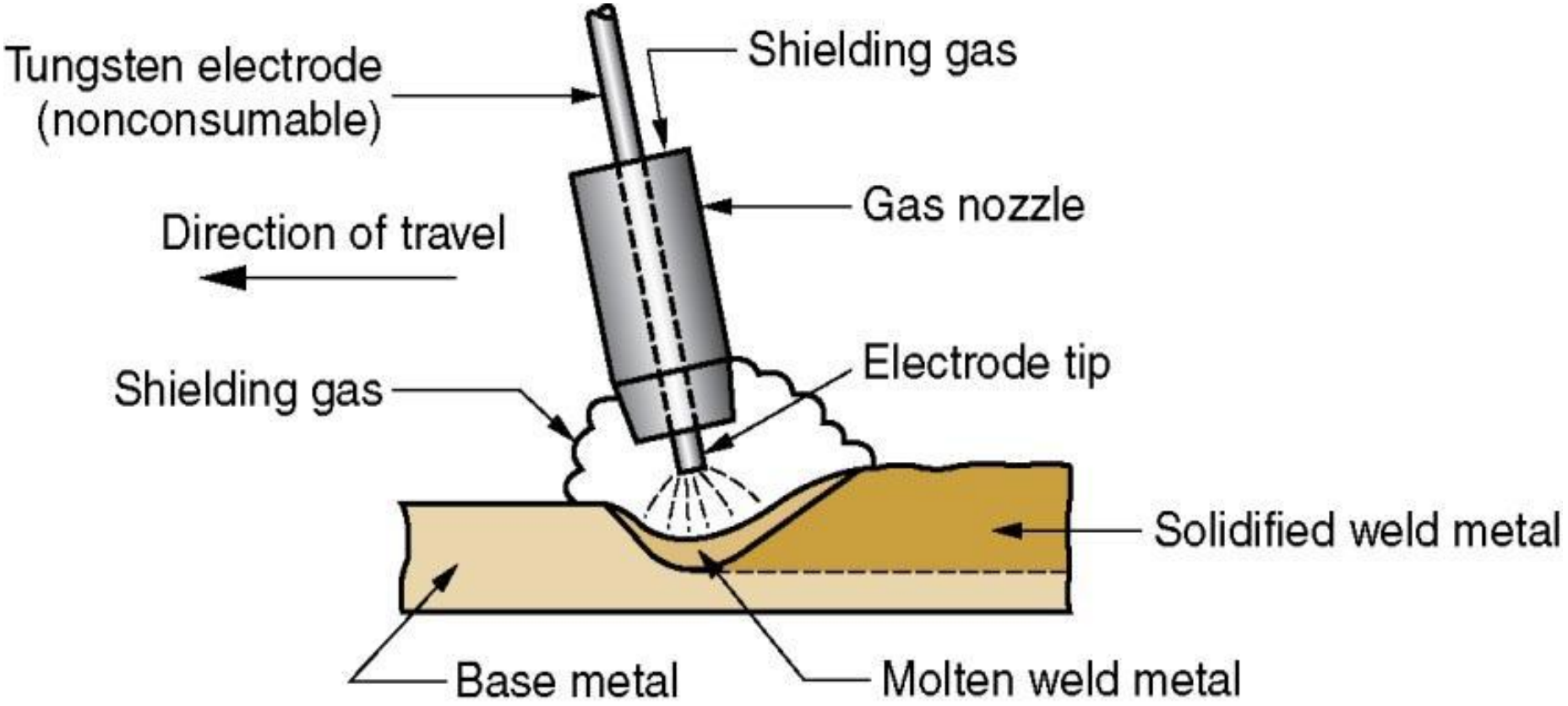
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Gas Tungsten Arc Welding (GTAW)

- Uses a non-consumable tungsten electrode and an inert gas for arc shielding.
- It is also called Tungsten Inert Gas (TIG) welding and known as Europe, called "WIG welding"
- It is used with or without a filler metal
- When filler metal used, it is added to weld pool from separate rod or wire.
- GTAW is applicable to nearly all metals in a wide range of stock thicknesses, mostly aluminium and stainless steel
- It can also be used for joining various combinations of dissimilar metals.

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Gas Tungsten Arc Welding (GTAW)



Advantages and disadvantages of gas tungsten arc welding

Advantages

- High quality welds for suitable applications
- No spatter because no filler metal through arc
- Little or no post-weld cleaning because no flux

Disadvantages

- Generally slower and more costly than consumable electrode AW processes

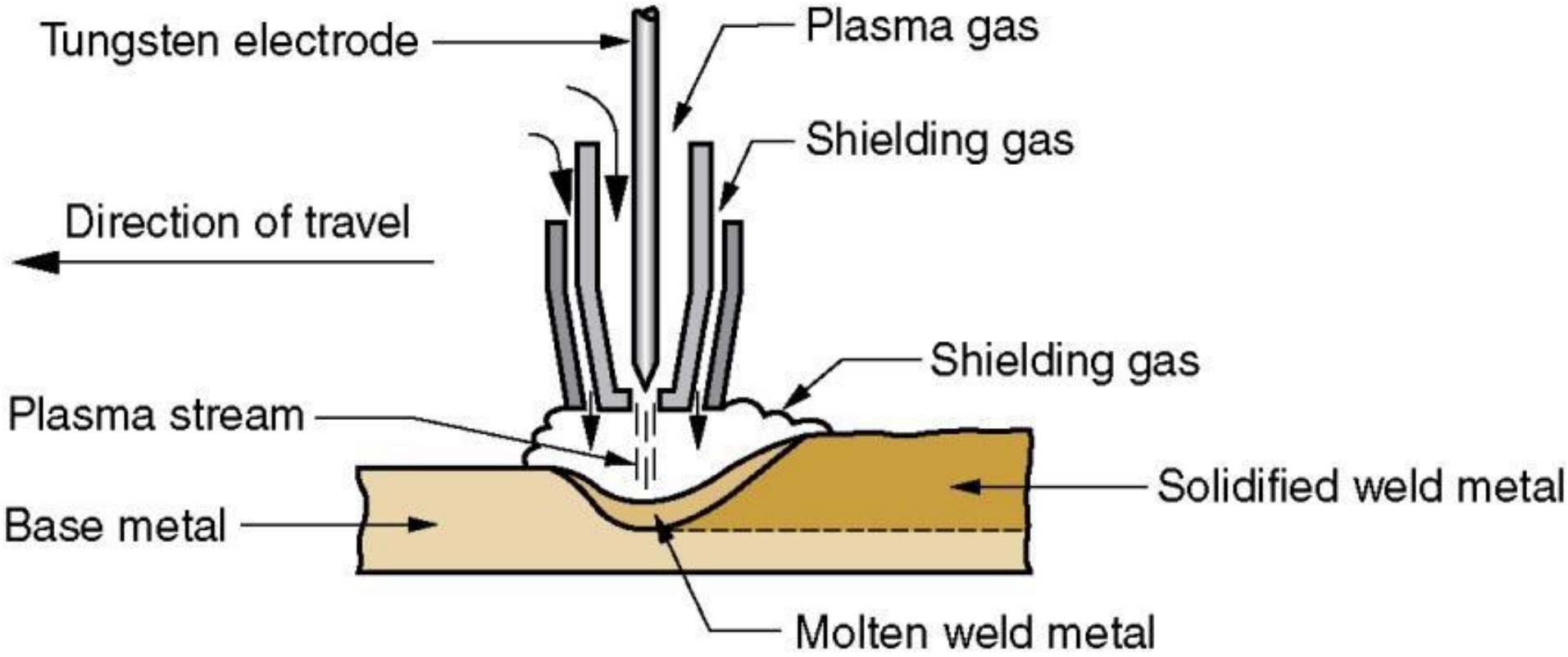
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Plasma Arc Welding (PAW)

- Special form of GTAW in which a constricted plasma arc is directed at weld area
- Tungsten electrode is contained in a nozzle that focuses a high velocity stream of inert gas (argon) into arc region to form a high velocity, intensely hot plasma arc stream
- Temperatures in PAW reach 28,000 ℃ (50,000℉), due to constriction of arc, producing a plasma jet of small diameter and very high energy density

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Plasma arc welding



Advantages and disadvantages of plasma arc welding

Advantages

- Good arc stability and excellent weld quality
- Better penetration control than other AW processes
- High travel speeds
- Can be used to weld almost any metals

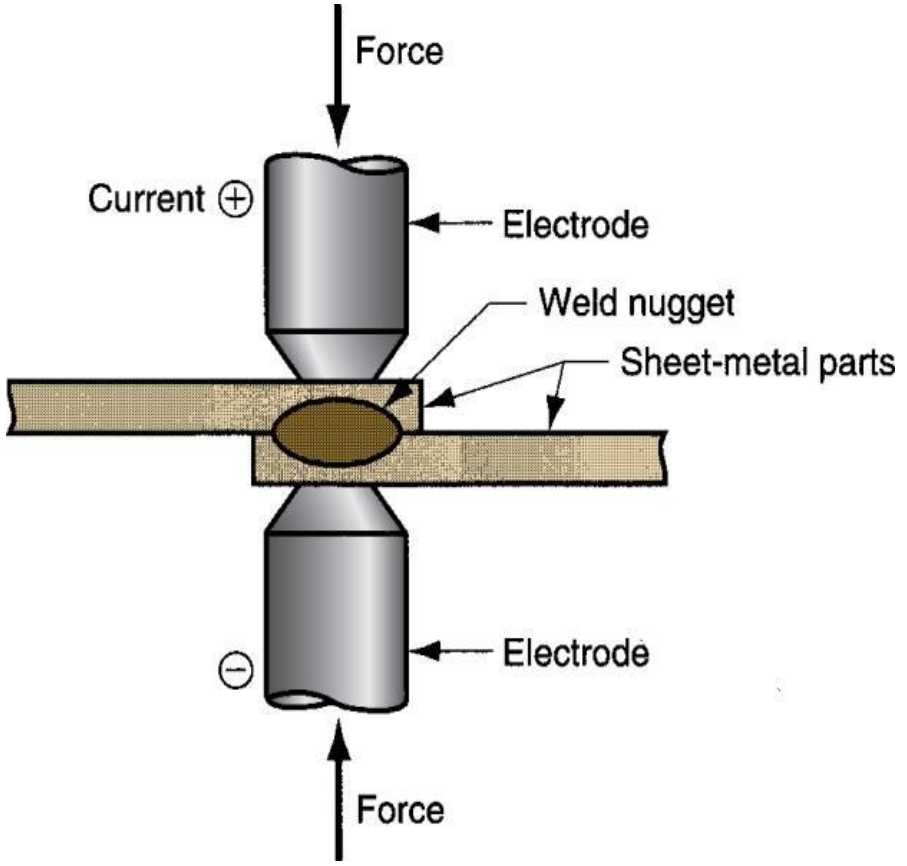
Disadvantages

- High equipment cost
- Larger torch size than other AW processes
- Tends to restrict access in some joints

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Resistance Welding (RW)

- A group of fusion welding processes that use a combination of heat and pressure to accomplish coalescence
- Heat generated by electrical resistance to current flow at junction to be welded
- Principal RW process is resistance spot welding (RSW)



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Advantages and disadvantages of resistance welding

Advantages

- No filler metal required
- High production rates possible
- Lends itself to mechanization and automation
- Lower operator skill level than for arc welding
- Good repeatability and reliability

Disadvantages

- High initial equipment cost
- Limited to lap joints for most RW processes

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Resistance Spot Welding (RSW)

- Resistance welding process in which fusion of faying surfaces of a lap joint is achieved at one location by opposing electrodes
- Used to join sheet metal parts
- Widely used in mass production of automobiles, metal furniture, appliances, and other products
- Typical car body has ~ 10,000 spot welds
- Annual production of automobiles in the world is measured in tens of millions of units

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Spot welding cycle

- (a) Spot welding cycle and (b) Plot of force and current cycle.

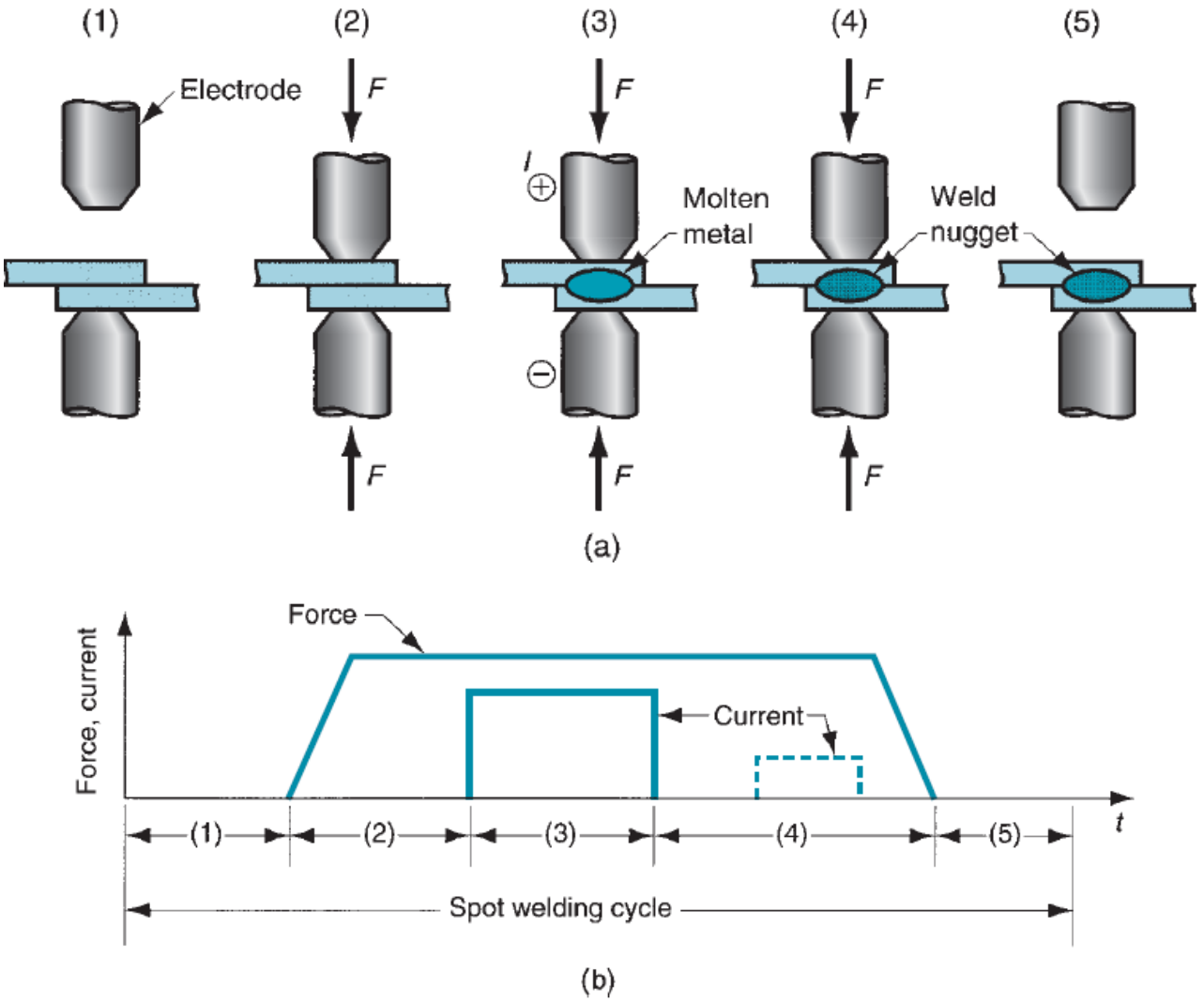
(1) Parts inserted between electrodes

(2) Electrodes closed

(3) Current switched on

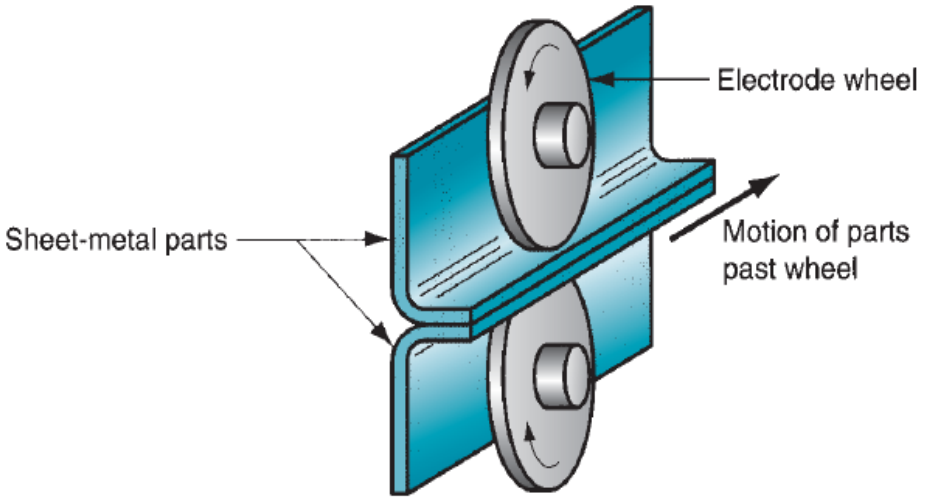
(4) current switched off

(5) Electrodes opened



Resistance Seam Welding (RSEW)

- Uses rotating wheel electrodes to produce a series of overlapping spot welds along lap joint
- Can produce air-tight joints



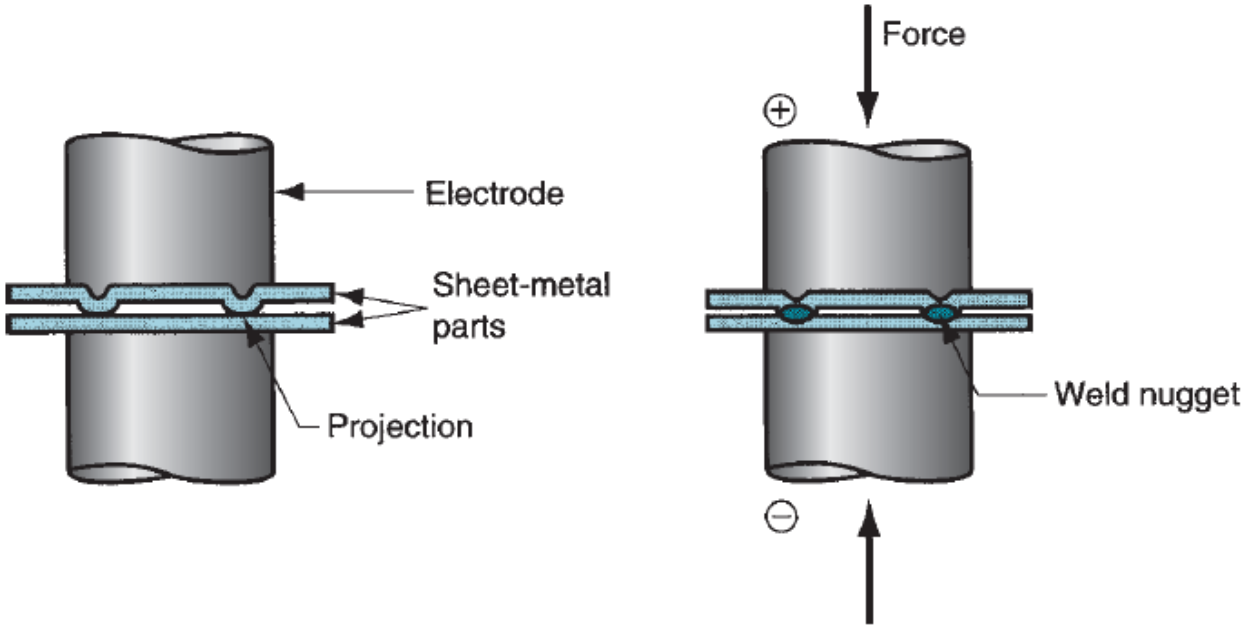
Applications

- Gasoline tanks
- Automobile mufflers
- Various sheet metal containers

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Resistance Projection Welding

- Resistance welding process in which coalescence occurs at one or more small contact points on the parts.
- Contact points are determined by design of parts to be joined.
- It may consist of projections, embossments, or localized intersections of parts.



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Oxyfuel Gas Welding (OFW)

- Oxyfuel gas welding (OFW) is the term used to describe the group of OFW operations that burn various fuels mixed with oxygen
- OFW employs several types of gases, which is the primary distinction among the members of this group
- Oxyfuel gas is also used in flame cutting torches to cut and separate metal plates and other parts
- Most important OFW process is oxyacetylene welding.

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Oxyacetylene Welding (OAW)

- Oxyacetylene welding is a fusion welding process performed by a high temperature flame from combustion of acetylene and oxygen
- Flame is directed by a welding torch
- Filler metal is sometimes added
- Pressure is occasionally applied in OAW between the contacting part surfaces

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Application of Pressure Gas Welding

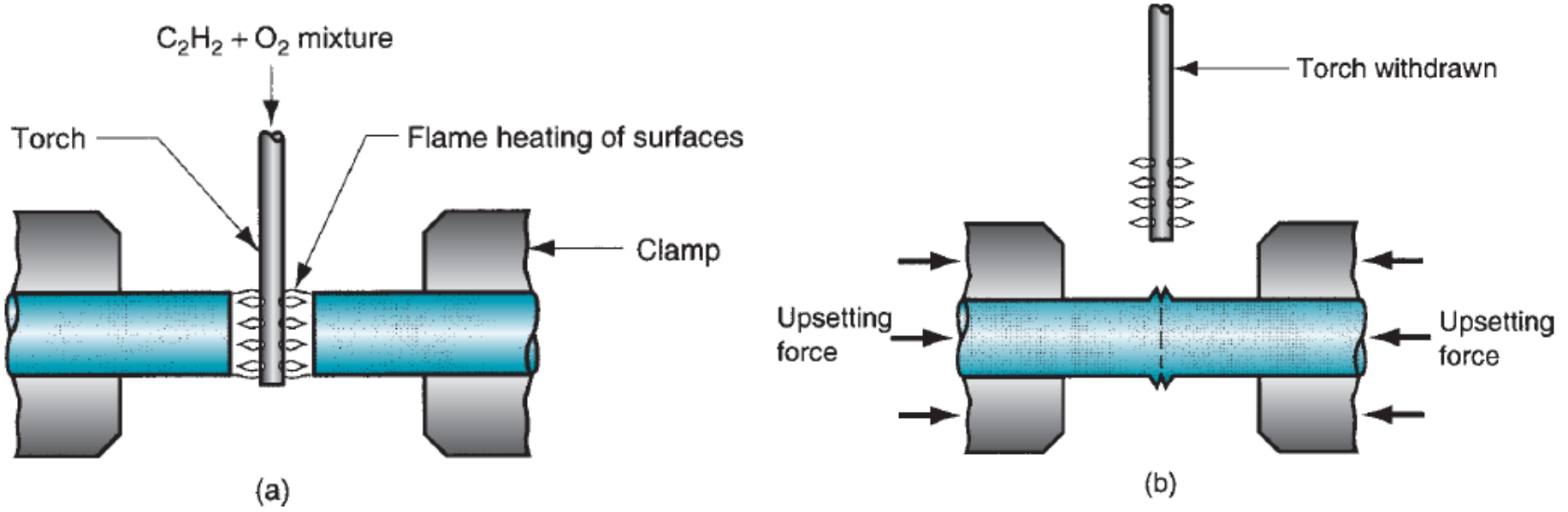
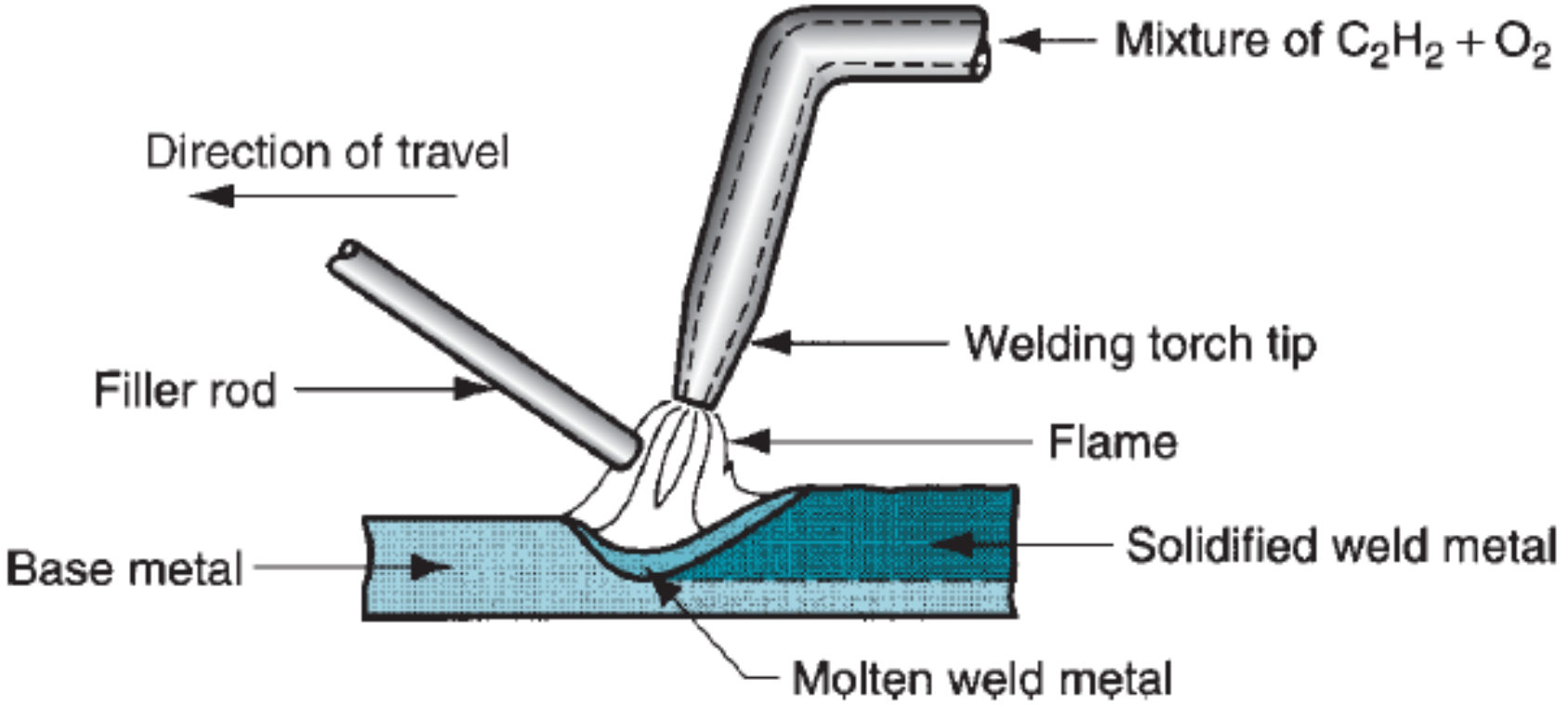


Fig. (a) Heating of the two parts (b) applying pressure to form the weld

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Oxyacetylene Welding Process



Acetylene (C_2H_2)

- Most popular fuel among OFW group because it is capable of higher temperatures than any other
- Up to 3480°C (6300°F)
- Two stage reaction of acetylene and oxygen:
  - ✓ First stage reaction (inner cone of flame)

$$C_2H_2 + O_2 \rightarrow 2CO + H_2 + \text{heat}$$

  - ✓ Second stage reaction (outer envelope)

$$2CO + H_2 + 1.5O_2 \rightarrow 2CO_2 + H_2O + \text{heat}$$

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Oxyacetylene torch

- Maximum temperature reached at tip of inner cone, while outer envelope spreads out and shields work surface from atmosphere
- Shown below is neutral flame of oxyacetylene torch indicating temperatures achieved.

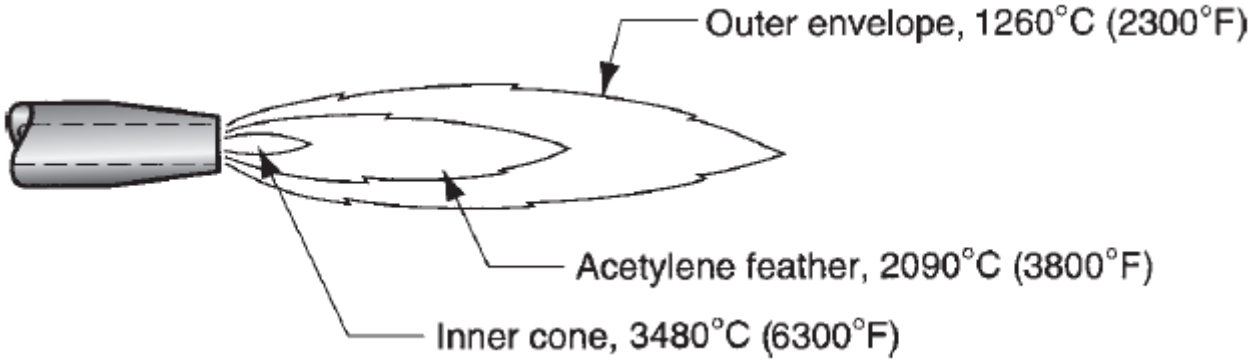


Fig. The neutral flame from an oxyacetylene torch, indicating temperatures achieved.

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Safety Issue in Oxyacetylene welding

- Together, acetylene and oxygen are highly flammable
- C_2H_2 is colourless and odourless. It is therefore processed to have characteristic garlic odour.
- C_2H_2 is physically unstable at pressures much above 1 atm.
- Different screw threads are standard on C_2H_2 and O_2 cylinders and hoses to avoid accidental connection of wrong gases.

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Alternative gases for oxyfuel gas welding

- Methylacetylene-Propadiene (MAPP)
- Hydrogen
- Propylene
- Propane
- Natural Gas

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Other Fusion Welding Processes

- There are other fusion welding processes that cannot be classified as arc, resistance, or oxyfuel welding
- These welding processes use unique technologies to develop heat for melting
- They have unique applications
- These processes include:
  - ✓ Electron beam welding
  - ✓ Laser beam welding
  - ✓ Electroslag welding
  - ✓ Thermit welding

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Electron Beam Welding (EBW)

- Fusion welding process in which heat for welding is provided by a highly-focused, high-intensity stream of electrons striking work surface
- Electron beam gun operates at:
  - ✓ High voltage (e.g., 10 to 150 kV typical) to accelerate electrons
  - ✓ Beam currents are low (milliamps)
- Power in electron beam welding is not exceptional, but power density is very high.

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Three Vacuum Levels in Electron Beam Welding

1. **High-vacuum welding** – welding in same vacuum chamber as beam generation to produce highest quality weld
2. **Medium-vacuum welding** – welding in separate chamber but partial vacuum reduces pump-down time
3. **Non-vacuum welding** – welding done at or near atmospheric pressure, with work positioned close to electron beam generator - requires vacuum divider to separate work from beam generator

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Advantages and Disadvantages of Electron Beam Welding

Advantages	Disadvantages
High-quality welds, deep and narrow profiles	High equipment cost
Limited heat affected zone, low thermal distortion	Precise joint preparation & alignment required
No flux or shielding gases needed	Vacuum chamber required
	Safety concern: Electronic beam welding generates x-rays

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Laser Beam Welding

- It is a fusion welding process in which coalescence is achieved by energy of a highly concentrated, coherent light beam focused on joint
- Laser beam welding is normally performed with shielding gases to prevent oxidation
- Filler metal not usually added
- It has high power density in small area
- Laser beam welding is often used for small parts

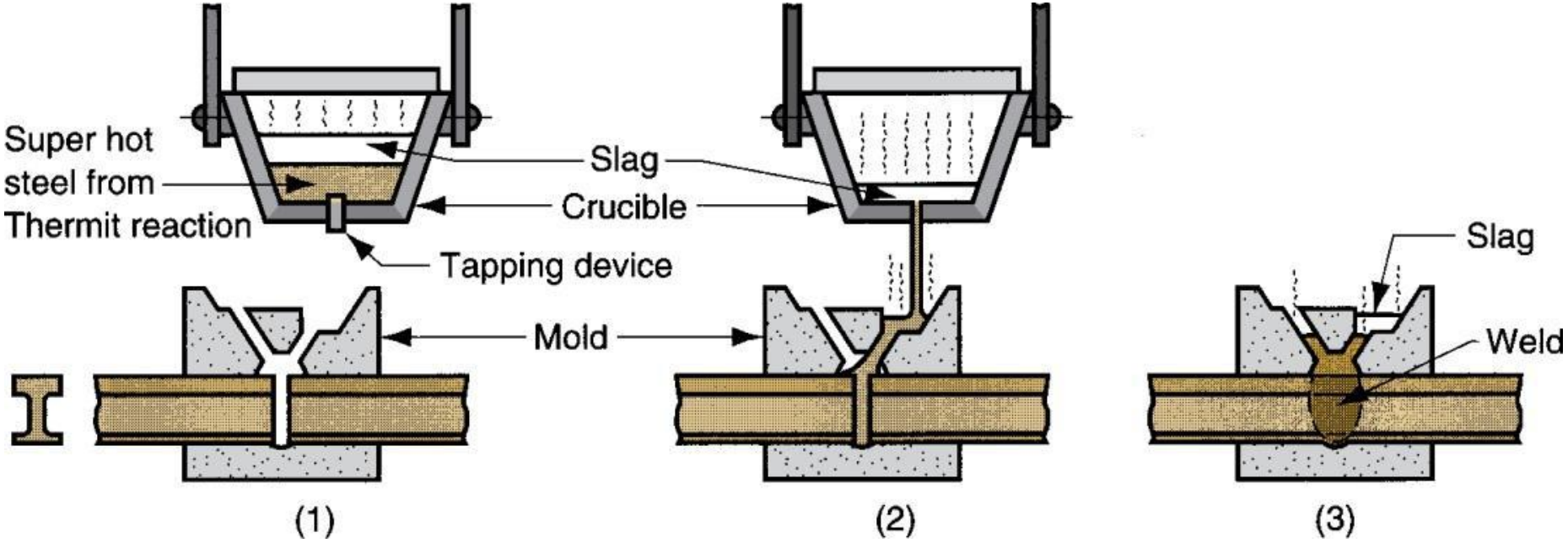
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Thermit Welding

- It's a fusion welding process in which heat for coalescence is produced by superheated molten metal from the chemical reaction of thermite
- *Thermite* = mixture of Al and Fe3O4 fine powders that produce an exothermic reaction when ignited
- Also used for incendiary bombs
- Filler metal obtained from liquid metal
- Process used for joining, but has more in common with casting than welding

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Thermit Welding



(1) Thermit ignited
(2) Crucible tapped, superheated metal flows into mold
(3) Metal solidifies to produce weld joint

Applications of Thermit Welding

- Joining of railroad rails
- Repair of cracks in large steel castings and forgings
- Weld surface is often smooth enough that no finishing is required.

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Solid state welding

- Coalescence of part surfaces is achieved by pressure alone, or heat and pressure.
- If both heat and pressure are used, heat is not enough to melt work surfaces.
- For some solid state welding processes, time is also a factor
- No filler metal is added
- Each solid state welding process has its own way of creating a bond at the faying surfaces.

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Advantages of solid state welding over fusion welding processes

- If no melting, then no heat affected zone, so metal around joint retains original properties
- Many solid state welding processes produce welded joints that bond the entire contact interface between two parts rather than at distinct spots or seams
- Some solid state welding processes can be used to bond dissimilar metals, without concerns about relative melting points, thermal expansions, and other problems that arise in fusion welding.

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Solid State Welding Processes

- Forge welding
- Cold welding
- Roll welding
- Hot pressure welding
- Diffusion welding
- Explosion welding
- Friction welding
- Ultrasonic welding

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Forge Welding

- It's a welding process in which components to be joined are heated to hot working temperature range and then forged together by hammering or similar means.
- It has historic significance in development of manufacturing technology.
- It is of minor commercial importance today except for its variants.

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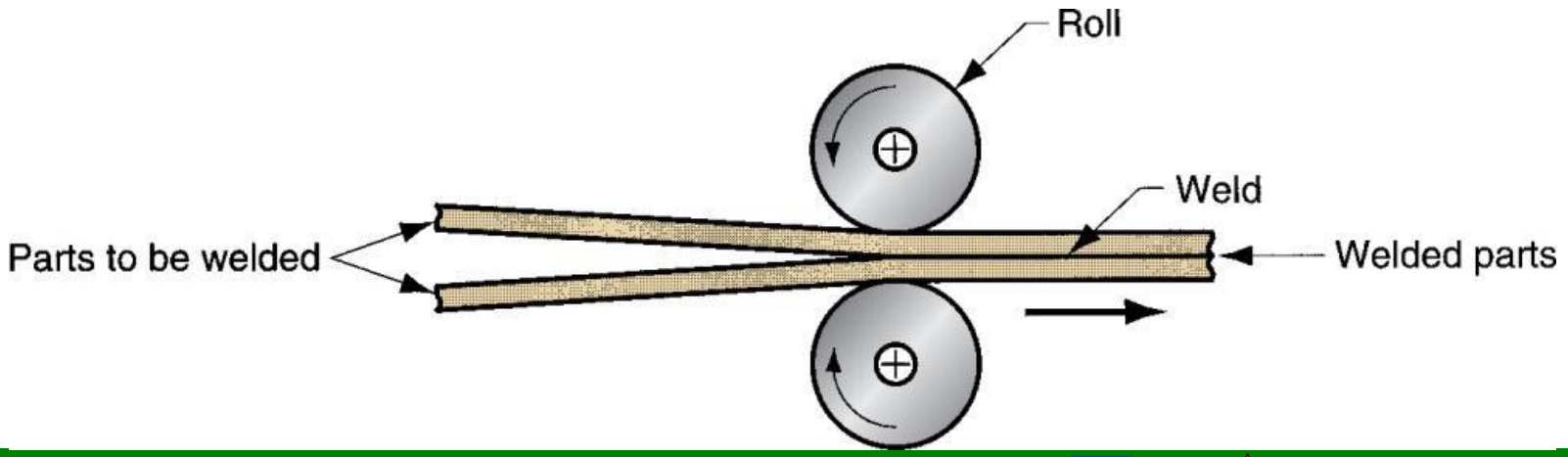
Cold Welding

- It is a solid state welding process done by applying high pressure between clean contacting surfaces at room temperature
- Cleaning is usually done by degreasing and wire brushing immediately before joining
- No heat is applied, but deformation raises work temperature
- At least one of the metals, preferably both, must be very ductile
- Soft aluminium and copper suited to cold welding
- **Applications:** Electrical connections

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Roll Welding

- It is a solid state welding process in which pressure sufficient to cause coalescence is applied by means of rolls, either with or without external heat
- Forge welding or cold welding varies, depending on whether heating of workparts is done prior to process
- If no external heat is applied, the process is called **cold roll welding**
- If heat is supplied, the process is **hot roll welding**



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Diffusion Welding (DFW)

- A solid state welding process that uses heat and pressure, usually in a controlled atmosphere, with sufficient time for diffusion and coalescence to occur.
- Temperatures $\leq 0.5\ T_m$
- Plastic deformation at surfaces is minimal.
- Primary coalescence mechanism is solid state diffusion.
- **Limitation:** time required for diffusion can range from seconds to hours.

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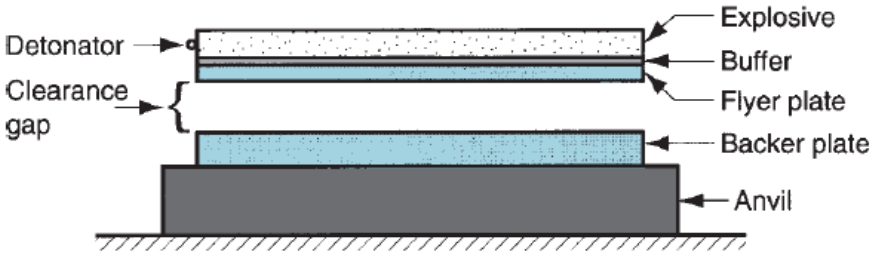
Applications of Diffusion welding

- Joining of high-strength and refractory metals in aerospace and nuclear industries.
- Can be used to join either similar and dissimilar metals.
- For joining dissimilar metals, a filler layer of different metal is often sandwiched between base metals to promote diffusion.

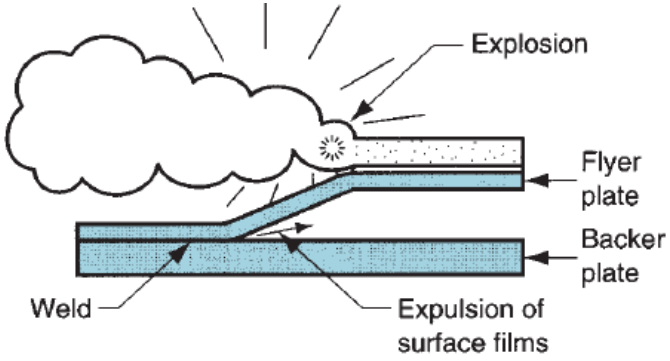
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Explosion Welding

- Solid state welding process in which rapid coalescence of two metallic surfaces is caused by the energy of a detonated explosive
- No filler metal used
- No external heat applied
- No diffusion occurs - time is too short
- Bonding is metallurgical, combined with mechanical interlocking that results from a rippled or wavy interface between the metals.
- Commonly used to bond two dissimilar metals, e.g., to clad one metal on top of a base metal over large areas



(1)



(2)

Friction welding

- A solid state welding process in which coalescence is achieved by frictional heat combined with pressure
- When properly carried out, no melting occurs at faying surfaces
- No filler metal, flux, or shielding gases normally used
- Can be used to join dissimilar metals
- Widely used commercial process, amenable to automation and mass production.

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Friction welding

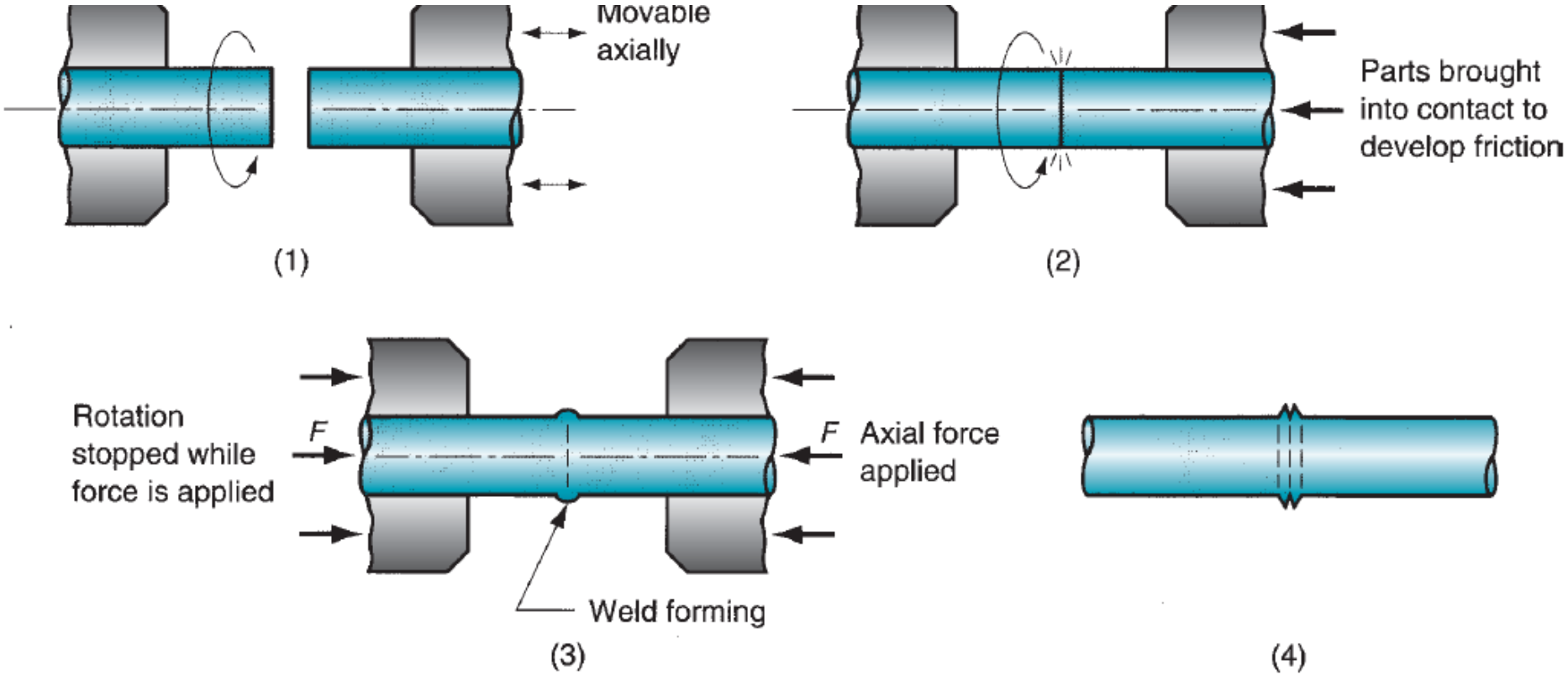


FIGURE Friction welding (FRW): (1) rotating part, no contact; (2) parts brought into contact to generate friction heat; (3) rotation stopped and axial pressure applied; and (4) weld created.

Applications and Limitations of Friction Welding

Applications

- Shafts and tubular parts
- Industries: automotive, aircraft, farm equipment, petroleum and natural gas

Limitations

- At least one of the parts must be rotational
- Flash must usually be removed (extra operation)
- Upsetting reduces the part lengths (which must be taken into consideration in product design)

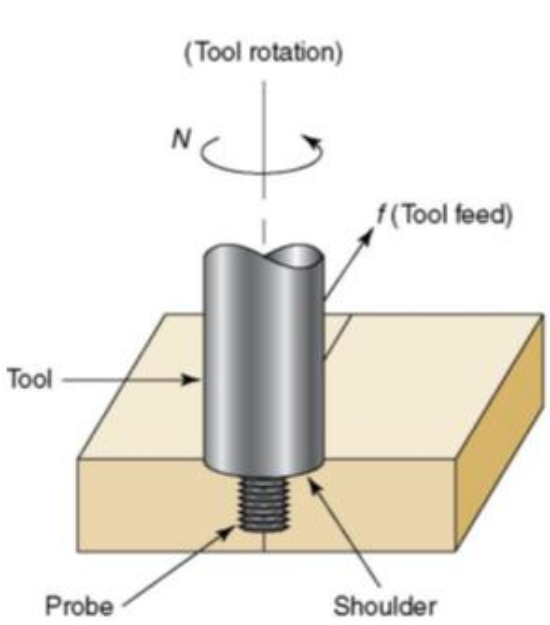
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Friction stir welding

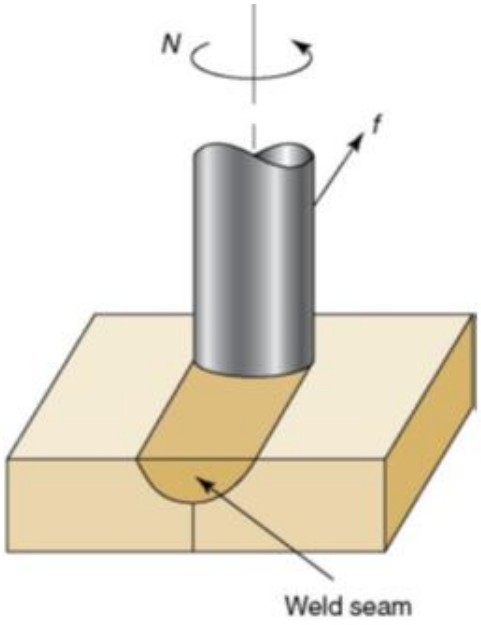
- Solid state welding process in which a rotating tool is fed along a joint line between two work-pieces, generating friction heat and mechanically stirring the metal to form the weld seam
- Distinguished from friction welding because heat is generated by a separate wear-resistant tool rather than the parts
- **Applications**: butt joints in large aluminium parts in aerospace, automotive, and shipbuilding

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Friction Stir Welding



Rotating tool just before entering work



Partially completed weld seam



Advantages and Disadvantages of Friction Stir Welding

Advantages

- Good mechanical properties of weld joint
- Avoids toxic fumes, warping, and shielding issues
- Little distortion or shrinkage
- Good weld appearance

Disadvantages

- An exit hole is produce when tool is withdrawn
- Heavy duty clamping of parts is required

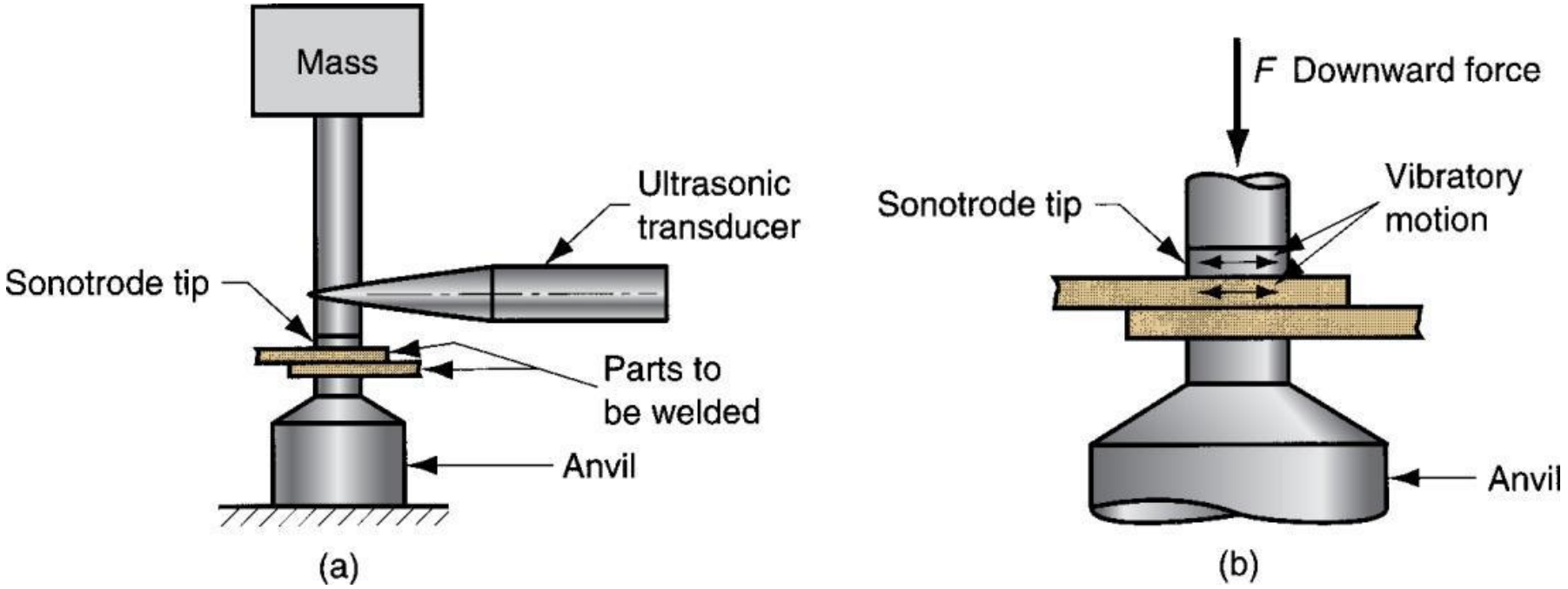
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Ultrasonic Welding

- Two components are held together, and oscillatory shear stresses of ultrasonic frequency are applied to interface to cause coalescence
- Oscillatory motion breaks down any surface films to allow intimate contact and strong metallurgical bonding between surfaces
- Temperatures are well below melting temperature
- No filler metals, fluxes, or shielding gases
- Generally limited to lap joints on soft materials

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Ultrasonic Welding



(a) General setup for a lap joint
(b) Close-up of weld area

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Applications of Ultrasonic Welding

- Wire terminations and splicing in electrical and electronics industry. Eliminates need for soldering
- Assembly of aluminium sheet metal panels
- Welding of tubes to sheets in solar panels
- Assembly of small parts in automotive industry

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Weld Quality

- It is concerned with obtaining an acceptable weld joint that is strong and absent of defects.
- It also involves methods of inspecting and testing the joint to assure its quality
- Weld quality issues include, residual stresses and distortion, welding defects and inspection and testing methods.

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Residual Stresses and Distortion

- Rapid heating and cooling in localized regions during FW result in thermal expansion and contraction that cause residual stresses
- These stresses, in turn, cause distortion and warpage
- Situation in welding is complicated because:
- Heating is very localized
- Melting of base metals in these regions
- Location of heating and melting is in motion (at least in AW)

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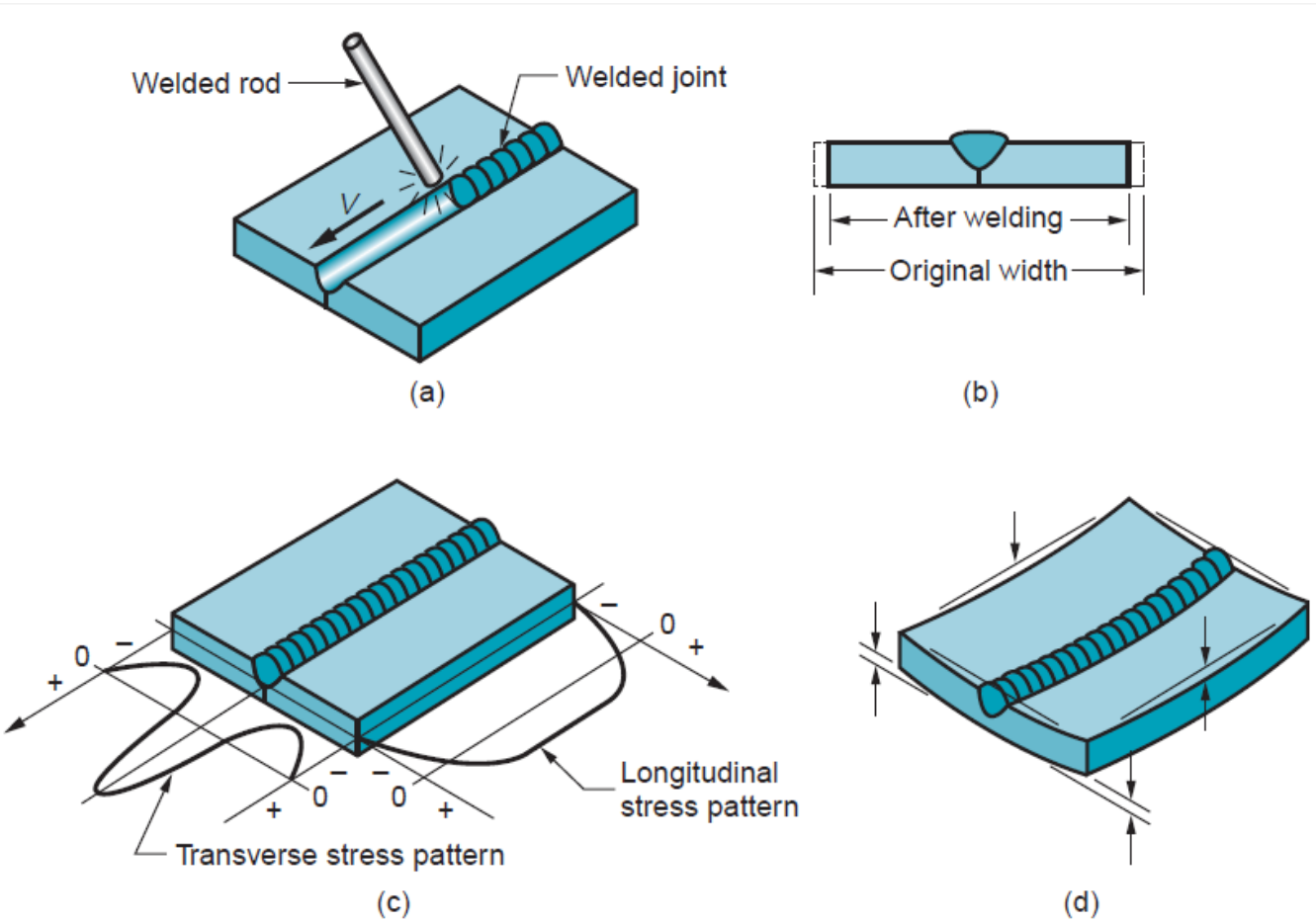
Residual Stresses and Distortion

a) Butt welding two plates

b) Shrinkage

c) Residual stress patterns

d) Likely warping of weldment



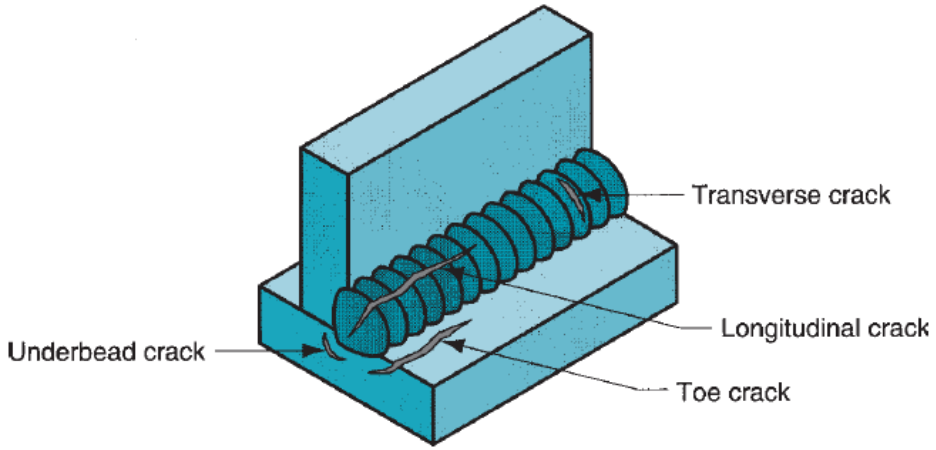
Defects in Welding

- Cracks
- Cavities
- Solid inclusions
- Imperfect shape or unacceptable contour
- Incomplete fusion
- Miscellaneous defects

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Welding Cracks

- Fracture-type interruptions either in weld or in base metal adjacent to weld
- Serious defect because it is a discontinuity in the metal that significantly reduces strength
- Caused by embrittlement or low ductility of weld and/or base metal combined with high restraint during contraction
- In general, this defect must be repaired
- Figure below shows forms of welding cracks



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Cavities

- Two cavity defect types, similar to defects found in castings exists in welding as follows:

  - ✓ Porosity - small voids in weld metal formed by gases entrapped during solidification. Caused by inclusion of atmospheric gases, sulphur in weld metal, or surface contaminants.
  - ✓ Shrinkage voids - cavities formed by shrinkage during solidification

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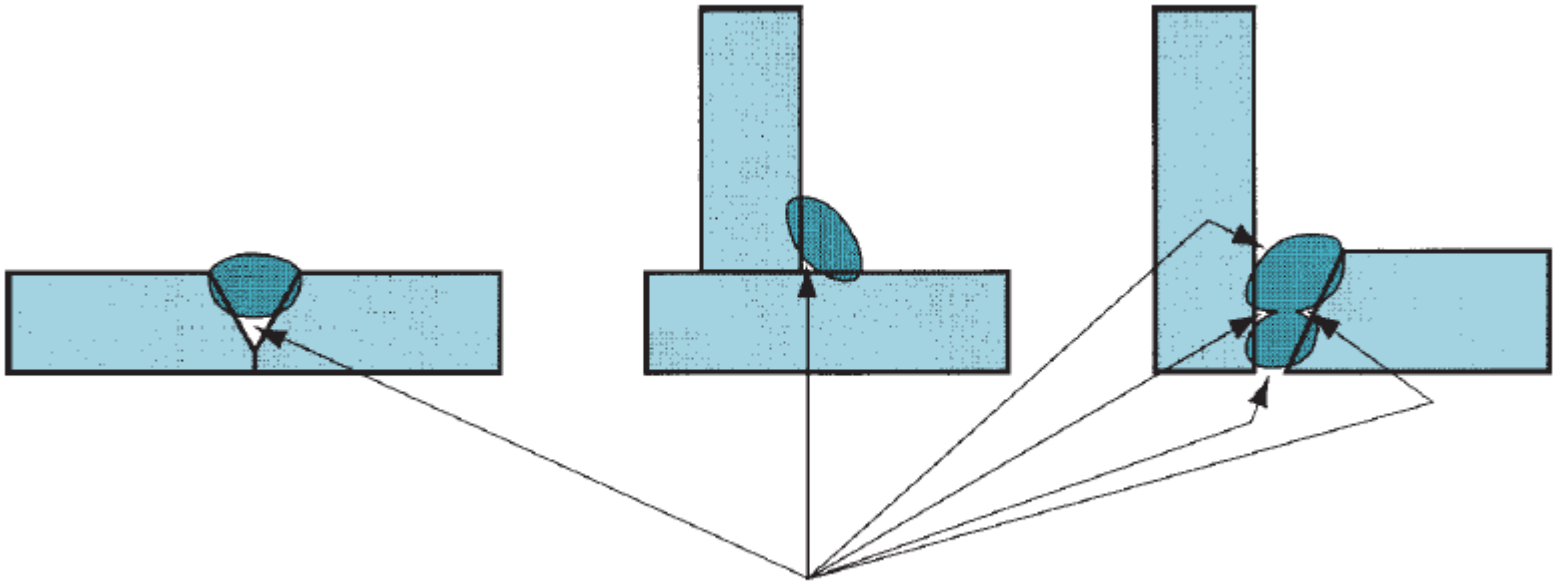
Solid Inclusions

- Non-metallic material entrapped in weld metal
- Most common form is slag inclusions generated during AW processes that use flux
- Instead of floating to top of weld pool, globules of slag become encased during solidification
- Other forms: metallic oxides that form during welding of certain metals such as aluminium, which normally has a surface coating of Al_2O_3

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Incomplete Fusion

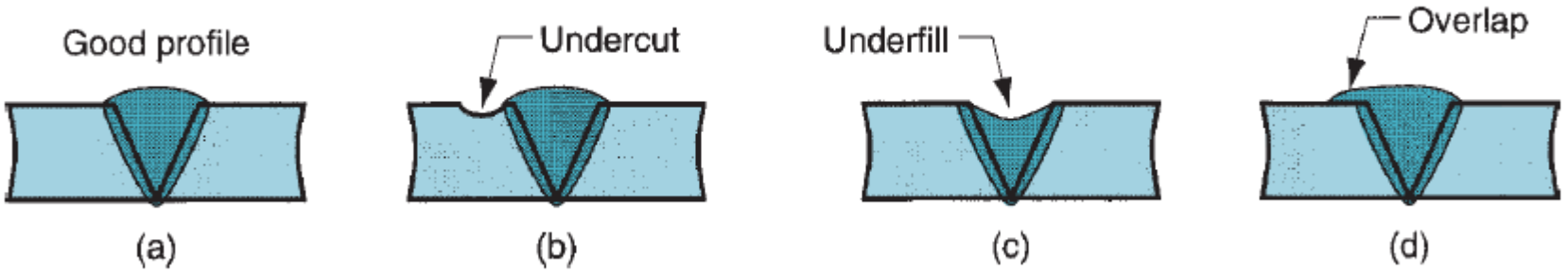
- It is also known as lack of fusion.
- It is simply a weld bead in which fusion has not occurred throughout the entire cross section of the joint.
- A related defect is ***lack of penetration,*** which means that fusion has not penetrated deeply enough into the root of the joint.



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Imperfect shape or unacceptable contour

- Weld should have a certain desired profile for maximum strength.
- This weld profile maximizes the strength of the welded joint and avoids incomplete fusion and lack of penetration.
- Some of the common defects in weld shape and contour are illustrated in Figure below



(a) Desired weld profile for single V-groove weld joint.
(b) undercut, in which a portion of the base metal part is melted away;
(c) underfill, a depression in the weld below the level of the adjacent base metal surface; and
(d) overlap, in which the weld metal spills beyond the joint onto the surface of the base part but no fusion occurs.

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Inspection and Testing Methods

- **Visual inspection:** It is the most widely used welding inspection method. Human inspector visually examines for conformance to dimensions, warpage, cracks, cavities, incomplete fusion, and other surface defects.
  - ✓ ***Limitations:*** Only surface defects are detectable. Welding inspector must decide if additional tests are needed
- Non-destructive evaluation:
- Destructive testing:

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Non-destructive Evaluation

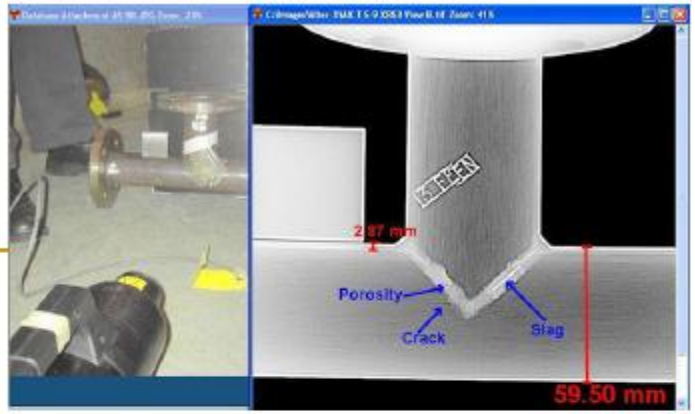
- **Ultrasonic testing** - high frequency sound waves through specimen to detect cracks and inclusions.
- **Radiographic testing** - x-rays or gamma radiation provide photograph of internal flaws.
- **Dye-penetrant and fluorescent-penetrant tests** - to detect small cracks and cavities at part surface.
- **Magnetic particle testing** – iron filings sprinkled on surface reveal subsurface defects by distorting magnetic field in part.

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Non-destructive Evaluation



Ultrasonic testing



Radiographic testing





Dye-penetrant



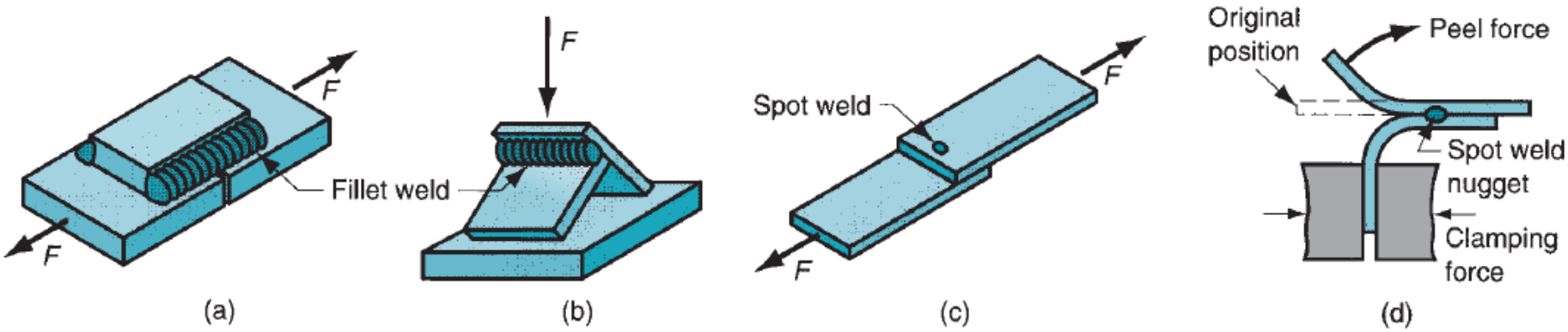
Magnetic particle testing

Destructive Testing

- In this type of testing, weld is destroyed either during testing or to prepare test specimen
- **Mechanical tests** - Purpose is similar to conventional testing methods such as tensile tests, shear tests, etc.
- **Metallurgical tests** - Preparation of metallurgical specimens (e.g., photomicrographs) of weldment to examine metallic structure, defects, extent and condition of heat affected zone, and similar phenomena.

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Mechanical Test Used in Welding



(a) tension–shear test of arc weldment
(b) fillet break test
(c) tension–shear test of spot weld
(d) peel test for spot weld.

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Weldability

- Capacity of a metal or combination of metals to be welded into a suitable structure, and for the resulting weld joint(s) to possess the required metallurgical properties to perform satisfactorily in intended service
- Good weldability characterized by:

✓ Ease with which welding is accomplished

✓ Absence of weld defects

✓ Strength, ductility, and toughness in welded joint

Factors of Weldability

- Some metals melt too easily; e.g., aluminium
- Metals with high thermal conductivity transfer heat away from weld, which causes problems; e.g., copper
- High thermal expansion and contraction in metal causes distortion problems
- Dissimilar metals pose problems in welding when their physical and/or mechanical properties are substantially different.
- Filler metal must be compatible with base metal(s).
- Moisture can result in porosity in fusion zone
- Oxides and other films on metal surfaces can prevent adequate contact and fusion

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Design Considerations in Welding

- **Design for welding** - Product should be designed from the start as a welded assembly, not as a casting or forging or other formed shape.
- **Minimum parts** - Welded assemblies should consist of fewest number of parts possible. Example: it is usually more cost efficient to perform simple bending operations on a part than to weld an assembly from flat plates and sheets.

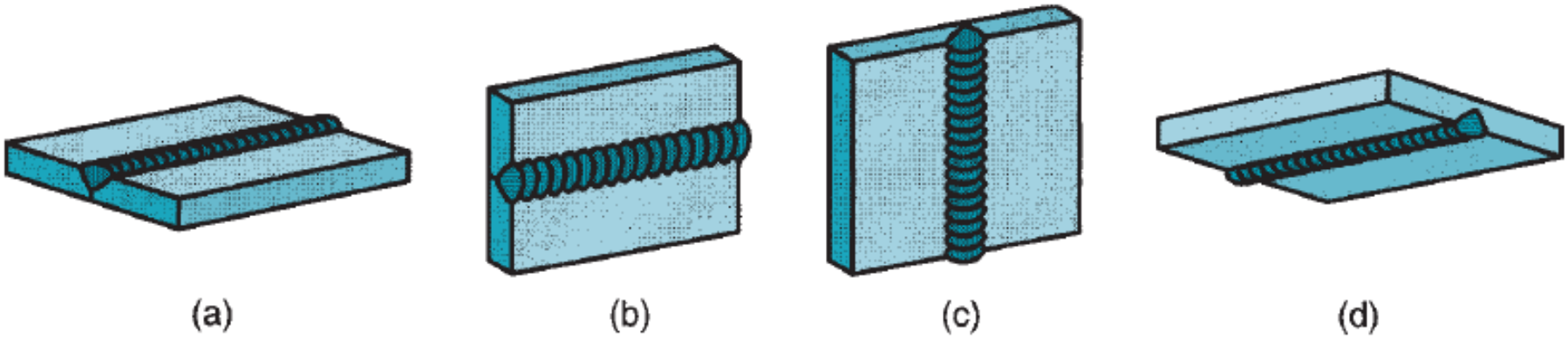
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Design Guidelines for Arc Welding

- Good fit-up of parts is need to maintain dimensional control and minimize distortion. Machining is sometimes required to achieve satisfactory fit-up.
- Assembly must allow access for welding gun to reach welding area.
- Design of assembly should allow flat welding to be performed as much as possible, since this is the fastest and most convenient welding position.

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Possible Welding Positions



(a) flat

(b) Horizontal

(c) Vertical

(d) Overhead (most difficult)

Guidelines for Resistance Spot Welding

- Low-carbon sheet steel up to 3.2 mm (0.125 in) is the ideal metal for resistance spot welding.
- Additional strength and stiffness can be obtained in large flat sheet metal components by spot welding reinforcing parts into them, or forming flanges and embossments into them.
- The spot-welded assembly must provide access for the electrodes to reach the welding area.
- Sufficient overlap of the sheet-metal parts is required for the electrode tip to make proper contact in spot welding.

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Thank You